ModelIt!

Student Activity Pack

The Food Web Puzzle

Remove One Piece and Watch Everything Change

Grade 05L2 | NGSS 5-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw a food chain with at least three organisms. Which one is the producer?

2 What do you think would happen if all the plants in an ecosystem died?

3 If wolves were removed from a forest, what would happen to the deer population?

4 What happens to a dead leaf on the forest floor over time?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Producer Population
- Primary Consumer Population
- Secondary Consumer Population
- Decomposer Activity
- Available Nutrients

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced Ecosystem

Normal producer population

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Remove Predators

Secondary Consumers set to 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Remove Producers

Producers locked at 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Sunlight Energy — The energy source that drives all producer growth

Tertiary Consumer — A top predator like an eagle or shark

Invasive Species — A non-native organism that disrupts the food web

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

What happens to an entire ecosystem when one species disappears? Use evidence from your simulation to support your answers.

1. **Why do ecologists say 'everything is connected' in an ecosystem?**

2. **What surprised you about removing predators from the system?**

3. **How do decomposers prevent dead matter from piling up?**

4. **What would happen if you added an invasive species to this model?**

5. **How does understanding food webs help us protect endangered species?**

8 STEM Challenge: Restore the Broken Ecosystem

ModelIt!

YOUR ENGINEERING MISSION

Design a plan to restore a damaged ecosystem where a key species has been lost.

THE SCENARIO

A lake ecosystem has lost its top predator (bass) due to pollution. Minnows (primary consumers) have exploded in population and are eating all the algae. Your team must restore balance.

Engineering Design Thinking

1. HOW WILL YOU REINTRODUCE OR REPLACE THE MISSING PREDATOR?
2. HOW LONG WILL IT TAKE FOR THE ECOSYSTEM TO REBALANCE?
3. WHAT INDICATORS WILL TELL YOU THE ECOSYSTEM IS RECOVERING?
4. WHAT OTHER SPECIES MIGHT BE AFFECTED BY YOUR RESTORATION PLAN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Ecologists and Conservation Biologists study food webs and ecosystem health to protect endangered species and restore damaged habitats. They earn $65,000-$100,000/year.

9 Post-Assessment

ModelIt!

Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the food web model, what happens when Secondary Consumers (predators) are removed from the system?

- **A.** Primary Consumers decrease because they have no food source
- **B.** Primary Consumers explode in population, overeat the Producers, and then Producers crash
- **C.** Producers increase because predators were eating them
- **D.** Nothing changes because predators are not important to the food web

2 What happens in the model when Producer Population is locked at 0%?

- **A.** Primary Consumers switch to eating decomposers
- **B.** Secondary Consumers increase because there is less competition
- **C.** The entire food web collapses because no energy enters the system
- **D.** Decomposer Activity increases to replace the producers

3 What role do decomposers play in completing the food web cycle?

- **A.** They eat producers and compete with primary consumers
- **B.** They break down dead organisms and release nutrients back into the soil for producers to use
- **C.** They create new energy from sunlight like producers do
- **D.** They have no connection to the rest of the food web

4 In a grassland, wolves eat rabbits and rabbits eat grass. If wolves are removed, what happens first?

- **A.** Grass grows faster because wolves were eating it
- **B.** Rabbit population increases rapidly because nothing is eating them
- **C.** Rabbit population decreases because wolves protected them
- **D.** Nothing changes because wolves only eat other predators

5 Why do ecologists say 'everything is connected' in an ecosystem?

- **A.** Because all organisms live in the same area
- **B.** Because changing one species population causes cascading effects on every other species through the food web
- **C.** Because all animals eat the same food
- **D.** Because all species were created at the same time